

Desk Research

Problem Statement

People who have difficulty understanding Dutch or complex medical language may receive healthcare information that they cannot easily understand or act on. This can make it harder for them to access and use healthcare information independently.

The problem is not necessarily that a person cannot read the information. Someone may be able to read the words but still struggle to understand medical terminology, instructions, or what action they are expected to take.

This is related to health literacy, which includes the ability to access, understand, evaluate and use health information and services.

Health Literacy in the Netherlands

A recent Nivel study investigated health literacy in the Netherlands. 11,441 people completed the questionnaire. The study found that 52.8% experienced difficulty with at least one aspect of finding, understanding, evaluating, and/or using health information. Evaluating information was particularly difficult. The type of information also mattered: participants found information about improving health especially difficult to find, understand, evaluate and/or use.

The study also found that people who struggle to make ends meet, are unemployed, or have a lower level of education experience more difficulties with health literacy.

What this tells us:

The problem is not only about translation.

People can experience difficulties with healthcare information for different reasons, including:

- language
- complex terminology
- difficulty evaluating information
- difficulty understanding what information means
- difficulty applying information to their situation

Language Barriers

Language can be an additional barrier for migrants and refugees.

WHO identifies linguistic barriers and low health literacy among the factors that can make it harder for refugees and migrants to access healthcare. WHO also identifies lack of information and difficulties communicating with healthcare services as relevant barriers.

WHO's review of communication barriers in European healthcare systems identifies several approaches that have been used to address these problems, including:

- interpretation
- translation of health information
- cultural mediation
- guidance and training for healthcare providers.

Why this matters for our project

This supports the idea that language and communication are already recognised barriers in healthcare, rather than being a problem we are assuming exists.

SDG 10 - Reduced Inequalities

The problem connects to SDG 10: Reduced Inequalities because access to understandable information can be more difficult for some groups than for others.

WHO describes health literacy as the ability to access, understand, evaluate and use health information and services. Limited health literacy can contribute to health inequalities.

In the Netherlands, a Nivel study found that 35% of adults have limited health literacy. Nivel also identifies groups that can experience greater difficulties with health literacy, including migrants, older people, people with lower education and people with limited digital skills.

This suggests that healthcare information is not equally easy for everyone to understand and use.

For our project, the relevant inequality is:

Unequal access to understandable healthcare information.

Users Group

People in the Netherlands who have difficulty understanding Dutch healthcare information because of language or medical terminology.

Primary: Patients with limited Dutch proficiency.

Secondary: Patients who understand Dutch but struggle with complex medical terminology.

Who is not the intended user: Healthcare workers or medical emergencies requiring immediate intervention.

Existing Solutions

There are already several ways of addressing language and communication barriers in healthcare, including:

- Professional interpreters
- Translated healthcare information
- Healthcare information in simpler language
- Cultural mediation
- Support from healthcare professionals

These approaches are important and already recognised by healthcare organisations and WHO.

However, our research suggests an opportunity to explore a more immediate form of support for a specific piece of healthcare information that a patient has received.

For example, a patient could receive a hospital message containing unfamiliar medical terminology and want to understand:

- What does this mean?
- What do I need to do?
- What information is important?

Ethical Considerations:

- Accuracy: AI can misunderstand or change important medical information.
- Medical misinformation: The AI should not invent information or change dosages, dates, numbers, warnings, or instructions.

- No medical advice: The tool should explain existing information, not diagnose, prescribe, or recommend treatment.
- Privacy: Healthcare information can contain sensitive personal data, so users should avoid entering unnecessary personal information.
- Transparency: Users should clearly know that the explanation is generated by AI.
- Human support: The tool should not replace doctors, pharmacists, or professional interpreters.
- Digital accessibility: The tool should be simple to use and consider people with limited digital skills or limited access to technology.
- Language and cultural differences: Direct translation may not always communicate the intended meaning, so explanations should focus on clarity rather than only word-for-word translation.
- Safety: If the information is unclear or involves an important medical decision, users should be directed to a healthcare professional.
- Main principle: Simplify the language without changing the important medical meaning.

Follow-up sheet (AI GENERATED)

Existing Healthcare Communication Solutions

There are already different ways to help people who have problems understanding healthcare information. We looked at some of the main solutions and their advantages and disadvantages.

1. Professional interpreters

What it does:

A professional interpreter helps a patient communicate with a healthcare professional in another language.

Advantages:

A real person can understand the situation and the context.

The interpreter can help during complex or sensitive conversations.

The patient can ask questions and get help during the conversation.

Disadvantages:

The patient needs access to an interpreter.

An interpreter may not always be available immediately.

It is mainly useful during communication with a healthcare professional, not necessarily when a patient is at home reading a message they already received.

2. Translated healthcare information

What it does:

Healthcare information can be translated into different languages so patients can read it in a language they understand.

Advantages:

Makes healthcare information available to more people.

Can help people who have limited knowledge of Dutch.

Disadvantages:

The exact information needs to be translated first.

A translated general information page may not answer questions about a specific letter or message that a patient has received.

3. Easy-language healthcare information

What it does:

Some healthcare organisations write information in simpler language. For example, Thuisarts aims to use clear Dutch and explains difficult words.

Advantages:

People can understand medical information more easily.

It helps people who struggle with complex language, even if they speak Dutch.

Disadvantages:

The information usually has to be written or adapted in advance.

It may not help with a specific hospital letter or doctor's message that was written in complex language.

4. Thuisarts translation

What it does:

Thuisarts allows users to translate selected text into another language. The translation uses DeepL, and the translated text can also be read aloud.

Advantages:

Easy to use on Thuisarts.

Supports different languages.

The text can also be read aloud.

Disadvantages:

It is mainly for information that is already on Thuisarts.

It is not mainly designed to explain any healthcare message that a patient receives from a hospital or doctor.

5. Translation apps

What it does:

Apps such as Google Translate and DeepL can quickly translate words, sentences and longer texts.

Advantages:

Fast and easy to access.

Can be used in many different situations.

Supports many languages.

Disadvantages:

Translation does not always mean that the patient understands the medical meaning.

A translated medical sentence may still contain difficult medical terminology.

Dutch healthcare guidance recommends caution when using digital translation tools for healthcare communication.

6. Pharos communication resources

What it does:

Pharos provides resources for making healthcare communication easier to understand. It has a large collection of tested healthcare communication materials.

Advantages:

The materials are designed with accessibility in mind.

Some materials have been tested with people who have difficulty reading and writing.

The resources can help healthcare organisations create clearer communication.

Disadvantages:

It is mainly a collection of resources for creating understandable healthcare communication.

It is not a tool where a patient can simply paste any personal healthcare message and receive an immediate explanation.

What did we learn?

These existing solutions show that language and healthcare communication are already recognised problems. However, translation and explanation are not exactly the same thing.

A patient may receive a specific healthcare message and still ask:

What does this mean?

What do I need to do?

What information is important?

Possible gap to explore

Instead of trying to replace interpreters or existing healthcare websites, our project could explore whether an LLM can help explain a specific healthcare message in simpler language and, when needed, in the user's preferred language.

The important question is whether the AI can make the information easier to understand without changing the important meaning of the original text.

Sources

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